

COOLING OF ELECTRONICS:

Thermal Management Techniques For Optimal Design



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All electronic devices and circuitry generate excess heat and, thus, require thermal management to improve reliability and prevent premature failure

Efficiency of an electronic device is inversely proportional to its temperature. A rise in temperature leads to a subsequent drop in performance. High-performance electronic components generate heat loads that create excessive junction temperatures, compromise component life-times and can lead to the premature failure of electronic devices.

As a result, there is a need to lower the temperature of and/or remove heat from electronic devices. This is a challenge often faced by the mobile device industry. One of the most challenging task is the ability to keep components cool, because overheating causes significant reduction in the operating life of a device and leads to device failure.

As electronic devices get smaller, engineers and designers are faced with the growing challenge of keeping up with the need to optimise processing speed within a shrinking form factor. Faster processors necessitate increased power consumption, which generates heat. A smaller form factor necessitates greater miniaturisation of the implements used to disperse that heat. These considerations are forcing engineers and designers to think in terms of systemic solutions in which every consideration in a device's power equation is examined for greatest optimisation.

In general, requirements of the power equation demand that heat dissipation must be proportional to the power dissipation of

a given device. Power dissipation is the amount of electricity wasted by a device (power dissipation is dependent on capacitance of logic elements, operating voltage swing and operating frequency).

Three major challenges to the problem of managing power

dissipation in today's small-footprint electronic devices are dense printed circuit boards (PCBs), increased heat density in integrated circuit (IC) chips, size of devices and mobility desired by users.

Most electronic components are low power and produce negligible amounts of heat in their operation. Some devices (power transistors, CPUs and power diodes), however, produce a significant amount of heat. Measures may be necessary to prolong their working life and increase reliability.

Let us consider a heat-producing electronic component in isolation. During operation, its temperature will rise until the heat produced within the device becomes equal to the heat lost to the surroundings, and the device has reached equilibrium.

Heat is generated due to the resistance offered by the material for the flow of electrons within it. It is given by the famous Joule's law of heating due to current flow as $H = I^2 R t$ (where, H is heat generated, I is current flowing, R is resistance offered by the material and t is time for which current is allowed to flow in the material). Resistance offered by the material for current flow is the main factor responsible for heat generation (if I and t are taken to be controlled). Lesser the resistance, greater the conductivity, and, thus, low heat generation. Hence, there are simultaneous efforts going on towards the research on high-conductivity materials.

Thermal management

Managing the heat generated by components has always been an important element of electronic design. It is even truer today, as power densities continue to increase and individual components such as microprocessors continue to draw more power to meet the demanding functional and computational requirements of current high-powered applications.

To ensure proper performance and reliability, excess heat must be transported away from critical components and system hot spots, and then dissipated into the ambi-

Fig. 1: Heat in electronics (Credit: www.ansys-blog.com)

